

	HINDUSTAN SHIPYARD LIMITED (A Govt. of India Undertaking) Ministry of Defence Gandhigram, Visakhapatnam - 530005	Electrical Design	REV: 002
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SHIP BUILDER
HINDUSTAN SHIPYARD LIMITED
 VISAKHAPATNAM- 530005.

POTS FOR MAIN BROADCAST / SOUND REPRODUCING EQUIPMENT (MB/SRE)

Project	:	DIVING SUPPORT VESSEL VC 11190 – 91
CLIENT	:	IHQ MOD (N) / DSP
DOCUMENT No	:	HSL/DSV/EL/SOTR/013

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002	Updated as POTS	25.05.2021	R.Sreenu	B. Ravi Kumar
001	Incorporated IHQ MoD(N) comments letter dtd. 09 Aug 20	10.09.2020	R.Sreenu	B. Ravi Kumar
000	First Issue to IHQ of MoD (N)	05.08.2020	R.Sreenu	B. Ravi Kumar
REV	DESCRIPTION	DATE	Prepared By:	AUTHORISED BY

ANNEXURE-1	Scope of Supply for MB/SRE	
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Chapter 1 - Preamble	
Clause No.	Description of Technical Specification
1.1	This document defines the technical requirements for design, development, manufacture, supply, testing, installation, setting to work, trials and onboard commissioning of Main Broadcast/ SRE complete with associated accessories and fitted on suitable mounts for 2Nos. of Diving Support Vessels (DSVs).
1.2	Scope: This Document outlines broad guidelines for hardware / software, interface, technical, functional, design, documentation & support requirement for MBSRE of IP-Based . It also specifies the applicable standards, environmental conditions, hardware/ system documents, user/maintainer training and various trials for the said systems.
1.2.1	Main Broadcast (MB) shall facilitate internal broadcasting and alarm annunciation, while the Sound Reproducing Equipment (SRE) shall facilitate crew entertainment onboard through common loudspeakers.
Chapter 2 - Functional Requirements	
2.1	MBSRE: MBSRE shall be a simplex system with Microphones, Announcement Points and Alarm Units placed at designated locations for relaying broadcasts & alarms in addition to facilitating crew entertainment onboard.
2.2	SRE: SRE sub-system shall be used for playing Radio, CDs, DVDs and USB when MB is not in operation. Radio, CD/DVD/USB player of standard and latest commercial make from reputed companies shall be supplied. There shall be a provision to connect an external music source through wired port.
2.3	Zone Selection Facility: MBSRE broadcasts shall be divided into two zones viz. General Broadcast (Zone-1) and Helo Deck Broadcast (Zone-2). Separate speakers with identical features shall be provided for both zones. The sound level of helo deck speakers shall be as high as 140dB. Zone Selection Facility Shall be provided at announcement points.
2.4	Priority: MBSRE System shall be able to accord priority to designated users enabling them

	to interrupt any other user. Order of priority shall be as follows: - First Level priority - Alarm Second Level priority - Announcement Point in Bridge Third Level priority - Announcement Point in OP's Room Fourth Level priority - Any of the other 06 Announcement Points Fifth Level priority - Paging from Auto Telephone Sixth Level priority – SRE																											
2.5	Alarms: MB shall be used for relaying alarm signals throughout the ship on highest priority. The system shall be able to generate 08 different types of distinguishable alarm tones. Required alarms with respective priority levels are listed below: <table><tr><th>Alarm Priority Level</th><th>Alarm Code</th><th>Alarm Types</th></tr><tr><td>1</td><td>A1</td><td>Man Overboard Alarm (through MOB push)</td></tr><tr><td>2</td><td>A2</td><td>Action Station Alarm</td></tr><tr><td>3</td><td>A3</td><td>Helicopter Crash on Deck Alarm</td></tr><tr><td>4</td><td>A4</td><td>Evacuation</td></tr><tr><td>5</td><td>A5</td><td>Call Captain</td></tr><tr><td>6</td><td>A6</td><td>Fire Alarm (from AFDS)</td></tr><tr><td>7</td><td>A7</td><td>Provision</td></tr><tr><td>8</td><td>A8</td><td>Provision</td></tr></table> Note: Flood Alarm for the Ship shall be catered through separate dedicated system. a) Helicopter Crash on Deck Alarm System: Shall confirm to the following: i) The helicopter crash on deck alarm shall be distinctive from shipboard alarms. ii) The helicopter crash on deck alarm shall be manually triggered and reset. iii) The helicopter crash on deck alarm shall be activated and mutable from the LSO compartment, DCHQ and the Bridge. b) General Alarm System: A general alarm system will be provided having push on the bridge console and bells in the important spaces as per Class rules and will be interfaced with RCS and MB SRE System. c) Action and Emergency Alarm: Main broadcast control box with action and emergency alarm in DCHQ and Section Bases.	Alarm Priority Level	Alarm Code	Alarm Types	1	A1	Man Overboard Alarm (through MOB push)	2	A2	Action Station Alarm	3	A3	Helicopter Crash on Deck Alarm	4	A4	Evacuation	5	A5	Call Captain	6	A6	Fire Alarm (from AFDS)	7	A7	Provision	8	A8	Provision
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2.5.1	Firm to elaborate alarm initiation and reset facility in the offer. It shall be possible to manually trigger and reset Helicopter Crash on Deck Alarm from the LSO compartment, DCHQ and Bridge.																											
2.6	Priority Configuration: The selection order of priority for the pre-amplifier input through logic circuit is to be made available. This priority for broadcast as well as alarms can be set by firm as per ship's requirement; the priority software is embedded and can be set at firm's factory or by replacing EPROM onboard.																											
	Chapter 3 - System Configuration																											

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3.1	MBSRE Rack with Amplifiers: All inputs/controls for Main Broadcast, interface with other systems, SRE (Radio Receiver, CD/DVD/USB) and all output amplifiers shall be supplied in two racks. Firm to note that limiting dimension for transportable sections for MBSRE equipment - doors size: 750mm X 1700mm and hatch size: 800mm X 800mm. Dimension, minimum clearance required for maintenance around the rack shall be indicated.										
3.1.1	Amplifiers required for the system shall be centrally located in a single rack. Amplifiers shall render adequate amplification to voice inputs before reproduction through speakers. <table><tr><td>ZONE</td><td>Area</td><td>Amplifier Rating (Approx.)</td></tr><tr><td>Z1</td><td>Gen. Broadcast</td><td rowspan="2">Total 12 x 240 Watt including standby</td></tr><tr><td>Z2</td><td>Helo Broadcast</td></tr></table>			ZONE	Area	Amplifier Rating (Approx.)	Z1	Gen. Broadcast	Total 12 x 240 Watt including standby	Z2	Helo Broadcast
ZONE	Area	Amplifier Rating (Approx.)									
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Z2	Helo Broadcast										
3.1.2	Additionally, a standard spare amplifier module shall be supplied and housed in the rack as standby. Switch-over to standby amplifier shall be automatic in case of failure of any main amplifier module.										
3.1.3	Grounding Arrangement: MBSRE cable screens shall be terminated at amplifier end only, other end shall be kept floating. Screens of two separate zones shall not be shorted. Firm shall make grounding arrangement at amplifier to avoid interference/ crosstalk.										
3.2	Loudspeaker: Common loudspeakers shall be used for MB and SRE in accommodation spaces, dining halls, galleys, workshops, offices, bathrooms, sickbay etc. However, when Main Broadcast or Alarm is initiated, SRE broadcast shall stop. For other locations onboard, only Main Broadcast shall be facilitated through the speakers.										
3.2.1	All Speakers shall have independent volume control. Volume control of the speakers shall be overridden during MB & alarm operation. All Speakers shall have minimum two cable glands for looping the cable. Following types of speakers shall be supplied as per the disposition: a) Speaker 1-4W Adjustable –IP44 b) Horn Speaker 1-15W Adjustable – IP56 c) Speaker 5W – EX Proof										
3.3	Handheld Microphone and Announcement Point: These shall be used for making announcements throughout the ship (Zone 1) or in Helo Zone (Zone 2) through the MBSRE speakers. The microphone shall be handheld type with an ergonomically positioned Press to Talk (PTT) switch, coiled cable and connector. It shall be equipped with built-in preamplifier to avoid cross talk and spurious pickups.										
3.3.1	Handheld microphone shall be connected to Announcement Point with zone selection facility for broadcast. Zone selection mechanism shall be in the form of a rotary switch/ push button arrangement with suitable indication.										
3.3.2	Announcement Point shall have a provision for stowing the handheld microphone. Alternately stowage hooks shall be supplied. However, for exposed deck locations, stowage box shall be supplied for stowing the microphone.										
3.4	Alarm Push Button/ Alarm Unit: Alarm units shall be used for initiating alarms to be relayed on the loudspeakers throughout the ship. These shall be separate or integrated along with Announcement points as per disposition indicated at SLD.										

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3.5	Headphone: Headphones shall be provided in Sick Bay/ Isolation Ward along with Headphone Connection Box (as required).
3.6	Junction Box: COTS type junction boxes with adequate terminals, suitable for marine use shall be provided.
3.7	Accessories: Accessories like stowage hooks/ boxes, clamps, socket boxes, base plates etc. shall be supplied as required.
3.8	Loud Hailer: The subunits of Loud Hailer shall be COTS type and suitable for marine use. IP56 speakers of 50W rating and a handheld microphone (in Bridge) shall be provided for the loud hailer. A dedicated 100W (approx.) amplifier shall be supplied for the loudhailer and housed in the central MBSRE rack.
	Chapter 4 - INTERFACE WITH OTHER SHIP SYSTEM
4.1	The firm shall provide all information required for integration of the equipment onboard. All components shall be compatible so that functional performance of the equipment is not degraded as a result of its integration. The firm shall interact with HSL and respond to queries from time to time for finalizing interface specifications/ protocols and resolving interface issues.
4.2	MBSRE interface with Auto Telephone Exchange: MBSRE system shall be interfaced with ship's Auto Telephone system for broadcasts through MBSRE speakers. Priority for broadcast shall be set by the MBSRE system. An Engaged indication shall be provided to the Telephone subscriber when Telephone broadcast is initiated during an ongoing broadcast. Further, during an ongoing Telephone broadcast through MB, if a higher priority broadcast is initiated, then the telephone broadcast shall be interrupted/ stopped through arrangement at MBSRE Rack. Interface data shall be shared by Auto Telephone vendor. MBSRE vendor shall make arrangements for compatibility towards seamless interface.
4.3	MBSRE interface with AFDS: Activation of fire alarm from potential free contact of the Automatic Fire Detection System (AFDS) shall be catered in the MBSRE system.
4.4	MBSRE interface with Man Over Board Push: Activation of MOB alarm from potential free contact of the Man Over Board Push Buttons (Part of Misc system) shall be catered in the MBSRE system.
4.5	Interface with ICC: Interface with microphone embedded in Integrated Captain's Chair (ICC) (part of IBS scope of supply) shall be carried out by the MBSRE during detailed design stage. PFC for each location that is LCC- 2 Nos and ICC – 1 No from the MB/SRE shall be provided.
4.6	Audio Recording and overlay on CCTV footage for LSO compartment: The MBSRE system shall have a provision of recording broadcasts relayed in LSO compartment in addition to announcements and alarms initiated from LSO.
4.7	MBSRE shall be interfaced with IP based Versatile Communication System through SIP protocol.
	Chapter 5 - Technical Requirement

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5.1	Power Supply: MBSRE system units shall be able to operate on 230V AC, 50Hz, 1Ph as primary supply (SMPS based) and backed up Secondary supply from 230V UPS. Provision for transient protection shall exist. Ship's power shall be made available from Main and Emergency sources for MBSRE system. Firm shall make suitable provisions in the systems to cater the power supplies available on-board as detailed below: 230V, 50Hz 1Ph/3Ph <table><tr><td rowspan="2">Voltage</td><td>Normal Voltage</td><td>230V</td></tr><tr><td>Load range tolerance</td><td>±8%</td></tr><tr><td rowspan="2"></td><td>Average Line to Line value of 3 phase system</td><td>±9%</td></tr><tr><td>Line to Line voltage of 1 phase of 3Ph system</td><td></td></tr><tr><td rowspan="2"></td><td>Maximum Unbalance</td><td>±2%</td></tr><tr><td>Maximum Modulation</td><td>±2%</td></tr><tr><td rowspan="2"></td><td>Transients (Excluding Spikes) (average line to line value for 3 phase system)</td><td></td></tr><tr><td>Frequent Transients</td><td>±9%</td></tr><tr><td rowspan="2"></td><td>High Limit</td><td>-13%</td></tr><tr><td>Low Limit</td><td>0.5 Sec</td></tr><tr><td></td><td>Recovery time</td><td></td></tr><tr><td rowspan="2">Wave Form</td><td>Maximum Individual Harmonic</td><td>3.5%</td></tr><tr><td>Maximum Total Harmonic</td><td>6%</td></tr><tr><td rowspan="2">Frequency</td><td>Nominal Frequency</td><td>50 Hz</td></tr><tr><td>Load range Tolerance</td><td>±2.5%</td></tr><tr><td rowspan="2"></td><td>Constant Load Tolerance</td><td>±0.5%</td></tr><tr><td>Modulation</td><td>±0.25%</td></tr></table>	Voltage	Normal Voltage	230V	Load range tolerance	±8%		Average Line to Line value of 3 phase system	±9%	Line to Line voltage of 1 phase of 3Ph system			Maximum Unbalance	±2%	Maximum Modulation	±2%		Transients (Excluding Spikes) (average line to line value for 3 phase system)		Frequent Transients	±9%		High Limit	-13%	Low Limit	0.5 Sec		Recovery time		Wave Form	Maximum Individual Harmonic	3.5%	Maximum Total Harmonic	6%	Frequency	Nominal Frequency	50 Hz	Load range Tolerance	±2.5%		Constant Load Tolerance	±0.5%	Modulation	±0.25%
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5.1.1	UPS: Supplier shall supply 230V UPS with SMF batteries for 1 Hr backup of adequate capacity in a single rack to meet the system requirement as per IRS class rules. Refer SLD of MB/SRE to arrive the capacity of UPS and batteries.																																											
5.2	Internal Wiring: Internal wiring of the equipment shall use LFH wires.																																											
5.3	Cable: System Cables are excluded from the firm's scope of supply except radio Receiver Cable. Cable specification and price per meter quotation for the same to be indicated in the offer. Any other special cables including intra unit cables (Fiber optic & CAT cables etc.,) shall be supplied by the firm. Shipyard will indicate cable OD & part no. for rest of the cables post placement of order. Inter unit cable diagram (Line diagram) specifying type of cable used to be provided by the firm along with the Binding Data.																																											
5.4	Cable Entry Glands: Cable entry glands shall conform to specification BS6121/ EN 624440-2013 for incoming and outgoing cables. Cable glands shall be fitted with the equipment, the gland size shall be suitable to the OD of the cable indicated by the Shipyard.																																											
5.5	Terminal: Bolted type/WAGO type terminal & crimped socket of electrolytic copper shall be provided for all incoming and outgoing cables. Adequate spare terminal strip shall be provided for provision of additional control and indication facilities. Adequate space shall be provided inside the equipment for bending and termination of incoming & outgoing cables.																																											
5.6	Earth Terminals: An external earth terminal of brass or stainless steel is to be provided for equipment.																																											

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5.7	Installation Material: One set of installation material per ship inclusive of special tools if any, special cables if any, plugs, adapters, mating flanges, sockets (with accessories), pins and cable lugs, vibration mounts etc. as required for satisfactory completion of system installation on board ship shall be supplied.
5.8	Components: All components shall be of Naval (Marine) approved type. Standard specification and grade of material of each component used shall be indicated in the binding drawings. Standard specification and grade of material of each component used shall be indicated in the binding drawings. Make and type of components are to be listed in the binding drawings. Indication lamps used shall be of LED type. The equipment shall be designed so that all components may be removed from the front of the cubicles with the front cover only removed. There shall be no hidden connection. All the components of the equipment shall have easy access for removal/ refit.
5.9	Enclosure: The enclosure shall be aluminum / steel / alloy / composite material having suitable arrangement for flush/ bulkhead mounting type installation on- board. Subunits shall be supplied with Ingress Protection – IP44/ IP56 as detailed in SLD for MBSRE. Enclosure protection for equipment shall be indicated in offer.
5.10	Lifting Arrangement: Equipment weighing more than 40 Kg shall be provided with collar eyebolts or suitable lifting lugs/ arrangements. If the eyebolts cannot remain in situ after the equipment has been installed in the ship, provision shall be made for their securing arrangement on the equipment.
5.11	Painting: Enclosure finish color shall be admiralty gray, semi gloss as per approved painting specifications/procedures. As per DQA(N) standards color code conform to IS5-COLOR CODE 632.
5.12	Tally and Diagram Plates: Tallies and diagram plates for all equipment shall be made of anodized aluminum alloy/SS. Size of tally plate and their letters shall conform to specification NES-723. 230V danger tally shall be provided on the equipment at a permanent place where required. Safety markings on main equipment and associated devices/units shall conform to NES 784.
5.13	EMI/EMC Standard: The equipment shall be capable of operating in the vicinity of other equipment / systems without causing or responding to undesirable electro-magnetic energy. The equipment design shall consider electromagnetic interference (EMI) coming from the external sources, generated from within its own enclosure and degrading the performance of the equipment. EMI may be reduced by judicious selection of components and use of good engineering practices like grounding, bonding and shielding. a) COTS equipment shall comply IEC 60945/IEC 60533 or equivalent standards. b) All equipment shall have the provision of grounding, bonding and shielding as per NECP 500 and MIL-STD-1310H.
5.14	Permissible Temperature Rise: The equipment must be designed so that the maximum permissible operating temperature of components is not exceeded during normal operating conditions in a maximum ambient temperature as specified in this specification.
5.15	Cooling: Experience in onboard use indicates a significant proportion of static conversion equipment failures occur as a result of overheating. Sufficient cooling arrangements by natural air convection shall be provided in the equipment in order to meet the permissible limit of temperature rise.

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5.16	Anti-Condensation Heaters: Approved type of non-luminous low temperature heaters operating at voltage 230V, 50Hz, 1 Phase, supply shall be fitted inside the devices at control panel to prevent condensation. The control of heater shall be provided in the system / equipment control panel. The Anti-Condensation Heater is to be switched on simultaneously when the equipment is in 'OFF' positioned arrangement for cutting off supply to the anti-condensation heaters when system is 'ON' position to be provided.
5.17	Reliability and Maintainability (R&M): (a) Mean Time Between failure (MTBF): The design of the equipment shall be such that the estimated mean time interval between failures shall be greater than 10000 hours. (b) Self Noise & vibration: Care is to be taken in the design and manufacture to ensure that the air borne noise and structure borne vibrations produced by the units/ devices, the self-induced air borne noise and vibration level shall meet ships stealth features.
5.18	Control and Protection: The following protection principles are to apply to all electrical and electronic equipment: a) Internal fault protection is to: (i) Prevent further damage within the equipment, subsequent to a fault (ii) Prevent damage to the converted power loads. (iii) Prevent the primary supply going out of specification b) External fault protection is to: (i) Prevent damage to the equipment, subsequent to a fault external to the equipment. (ii) Where practicable, include current limiting techniques to allow the equipment to work with a short circuit onto the output. (iii) Be defined in terms of conditions under which the equipment shall trip. c) Estimated and calculation of the effects of fault conditions are to include: (i) Stored energy in the load (e.g. resulting from shunt capacitors or motor loads) (ii) Stored energy in the equipment itself (e.g. in back-up batteries) d) The design of protective arrangements shall take into account thermal capacity of the components being protected and heat sink arrangements incorporated within equipment. The contribution to R&M figures of protective arrangements must be considered.
5.19	ESD Protection: The system design shall take into account adequate measures for Electro Static Discharge (ESD) control and protection at PCB/ module/ assembly and unit level. Each Electro Static Discharge sensitive part/assembly shall be duly marked with a symbol/ warning. The firm shall use ESD protective materials for handling, packaging storage and transport.
5.20	Environmental Stress Screening (ESS) Test: ESS test for electronic components/ sub-systems is required to be carried out for 100% components/ parts/ assemblies as per IHQMOD(N)
Chapter 6 – General	

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6.1	Ship's Data: The System shall be suitable for fitment on DSV class of ship with the following features:- a) Length Overall - 119.40m b) Length Between Perpendiculars – 110.70m c) Breath MLD – 22.80m d) Depth to Deck10 – 10.40m e) Design Draft – 6.00m f) Total compliment - 215
6.2	Classification Society: Vessel Shall be designed, built and outfitted as per applicable provisions and regulations as given in the Naval Rules of Indian Register of Shipping. The Class Notation for the Diving Support Vessel as follows. Swastika SUL "Diving Support Vessel", FDA, VIB(HGL), HELIDK, RAS(AB) Swastika IY, DP(2), EP, DSV Descriptive Note: "Submarine Rescue vessel", "Deck strengthened for 5t/m2", "Special Government Service".
6.3	Operating Condition: The system devices/equipment shall operate satisfactory in the marine environmental (coastal conditions) and other operating conditions enumerated in the succeeding paragraphs.
6.3.1	Seaway conditions: a. Roll : Max 20°with 10 sec period. b. Trim : Max 5° c. List : Max 20°from vertical. Pitch : Max 6°with 20 sec period.
6.3.2	Sea Worthiness: Main equipment along with its control equipment & accessories should meet sea worthiness criteria as mentioned below.The sea worthiness of the ship shall be as follows:- a. Sea worthy for transit on all headings upto Sea State 7 and survivable on best headings at Sea State 8. b. Capable of operating Helicopter in Sea State 6 on best heading. c. Capable of conducting Diving Operations upto Sea State 4. d. Capable of Dynamic Positioning operations upto Sea State 4.
6.3.3	Units and Language: Language shall be English. Metric Units shall be used.
6.3.4	Environment Condition: The equipment and the accessories including all integral and associated service system should be capable of continuous operation when the ship is in the following conditions: a. Ambient Conditions : b. Sea water temp : 1 deg.C to + 35 deg.C c. Max relative humidity : 100% at 38 deg C d. Salinity of water : upto 35 ppt

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	Max. Machinery room temp: Machinery room compartments maintained at ambient (33°C) + 5 in open condition and between 45 to 55°C in closed condition.
6.4	Commonality: The firm to mention the level of commonality achieved within the system at the following levels :- (a) Module/Sub- module level (b) PCB level (c) Component level
6.5	Weight Data Sheet: The firm to furnish Weight data sheet for Equipment detailing the following for the use of calculation for weight curve :- (a) Carcass weight (b) Total weight of equipment (c) Centre of gravity showing 3 axis coordinate
6.6	Security: The information about the ship and the requirements mentioned in the SOTR are not to be communicated to any third party without prior approval of IHQMOD(N).
6.7	Warranty: The Equipment with Associated controls/ instrumentation shall be guaranteed for stipulated performance for a time period mentioned as per STAC/Commercial Clause. The equipment supplied shall be warranted from defects arising due to the manufacturer/firm and performance for the said period and cover all the defects arising from malfunction through design faults, inappropriate material, bad production and non-conformance to specifications. Any expense on account of repair/ supply of spares against guarantee defects shall be borne by the firm.
6.7.1	Extension of warranty: Firm shall mandatorily indicate in the offer, the annual rate for extended warranty period, in the event of expiry of warranty period as detailed above along with the associated commercial terms and conditions if any.
6.8	Upgrades and Obsolescence / Product Support: Latest version of the equipment/ system at the time of tendering shall be provided. All upgrades and modifications thereafter in the hardware and software of the system need to be communicated to the Navy for the complete support life of 25 years from the date of commissioning of the equipment on board ship excluding electronic item for which the period is 10 years. Obsolescence of subsystem, components, parts and assemblies need to be communicated to Navy at least two year in advance with recommended remedial action. A maintenance contract may be offered for consideration and negotiation at the later date. Firms supplying imported equipment/systems are to ensure that product support and aftersales service in India are available for all assemblies/ modules/ software including third party supplies prior commencement of trials.
6.9	Safety standards: Equipment shall offer total safety to personnel from moving parts and other associated equipment from vibrations. Safety standards as per MIL STO 454/ equivalent shall be met.

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Chapter 7 - Spares, Tools and Test Equipment	
7.1	Onboard Spares : OEM recommended onboard spares for schedule servicing, maintenance & break down maintenance falling due within the first operation cycle for 02 years shall be provided for the equipment in ILMS Ver 2 format (Integrated Logistics Management System) indicating the item code number, quantity and item wise cost. The Spares consumed within the guarantee period to rectify / liquidated known defects shall be replaced free of cost. The supplier shall prepare the list of OBS (including onboard maintenance tools) based on the Reliability and Maintainability data and taking into account IN's maintenance philosophy, and shall forward to IHQ MOD (Navy) for vetting. The complete inventory of system parts including OBS & B&D spares is to be provided in INCAT (Indian Naval Catalogue of Inventory) compatible format in electronic media for ILMS (Integrated Logistics Management system) of Indian Navy, for the management of spares. The supplier shall provide both hard copy as well as soft copy.
7.2	Five Year Base and Depot (B&D) Spares. Base spares recommendation shall cover maintenance / overhaul requirements of 5 years including two refits shall be provided for the equipment in ILMS Ver 2 format. Recommendations for insurance holding of long lead items are also to be indicated. Itemized cost of B&D spares are to be forwarded with the main offer along with part identification list of spares. The list would contain price, description, Pattern no. and quantity fitted on each equipment in respect of various parts / components. The quotation of spares shall be valid for at least 18 months. The Buyer shall subsequently range and scale the B&D spares. The cost negotiation of B&D spares would be carried out by a committee constituted by the Buyer, wherein the seller's and sub-vendor's representative would be a member. The OEM shall undertake procurement of the ranged and scaled spares as per the ordering instructions issued by the IHQ MoD (N).
7.3	Installation & Commissioning Spares: The supplier's service of engineer shall supply/ carry one set of installation and commissioning spares along with the main equipment.
Chapter 8 - Binding Data , Design Drawing & Technical Documentation	
8.1	Binding Drawing: The firm shall submit approved and as made drawings sufficiently detailed to show the manner of construction and operation, the method of assembling and dismantling. The following binding drawing/documents in duplicate (both hardcopy & softcopy) shall be submitted to IRS for approval with a copy to shipyard within 4-6 weeks after placement of the order. (a) Installation drawings indicating overall dimensions, C.G., weight, maintenance envelope, IP rating, power ratings, vibration mount details etc of each unit. Detailed foundation drawing including bolting plan shall be

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	included. (b) Mounting arrangement drawings with installation instructions. (c) Heat Dissipation of individual units in compartment & in close loop ventilation system, as required for the system. Other interface requirement from ship viz. air, water, cooling etc to be indicated. (d) Material list Indicating Part No., Quantity, Material, Makers name etc. (e) Block diagram of all major assemblies showing interconnection between these assemblies as well as cables specifications. (f) Power supply scheme for the system with specific mention of UPS requirement. (g) Cable length limitation etc, if any. (h) Recommended arrangement of devices in nominated compartments. (i) Circuit diagram, wiring diagrams, wiring schedules, terminal connection and core connection details. (j) Details of standard and optional factory tests
8.2	Design Documents/Drawings: All associated design documentation, drawings and equipment list applicable must be in English language. All drawings and documents shall contain dimensions and other parameters in metric units (SI Units). The drawing and documentation shall be complete to enable installation on board. These shall cover all subassemblies and accessories of the equipment. The documents/drawings shall be provided in both hard copy form and soft copy in pdf format which could be copied on CD_ROM. Following draft documents shall be presented for approval of IHQMOD(N): a) Technical and operation Instruction Manual b) Onboard Maintenance Manual c) field and Depot Maintenance manual d) Operation Maintenance document e) Installation and testing specifications/manual f) Certified Test Data and report g) Parts and Tools Catalogue h) Installation drawings in reduced Size i) As Fitted Drawings j) Any Other Special Instructions for Installation and Preservation k) Conformation of Standards Specified l) Test Procedures and Documentation m) Heat Dissipation of Devices n) Storage, handling, transportation details etc. o) Details upto 2nd level maintenance for repair and troubleshooting. Component level maintenance details to be provided along with list of components. p) The software in the system shall comply with any of the established standards like MIL-498, IEEE 12207 etc. All Documents as per the standards related to software design, setting and maintenance shall be provided.

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8.3	Technical Documentation: Details of technical documentations to be supplied are as mentioned below: Separate manuals in respect of following for both equipment and system shall be provided in hard format:- (a) Operation and Specifications including onboard Repair and Maintenance. This should include information about the equipment (such as serial number, trial reports, etc) and the manufacturer (such as addresses, telephone and fax numbers, email, etc). (b) Repair Technical Document with detailed drawings (c) Complete Parts List			
	S.No	Description	No. of Copies	
	1	User Handbook (Including Technical Description, operation Instruction and Service log Book)	Two sets of documents shall be supplied in CD_ROM in addition to nine hard copies for each ship.	
	2	Technical Manual including Setting to work & test Procedures.		
	3	CPIL/PIL		
	4	Servicing and Maintenance Manual (field and depot)		
	5	Unpacking and Installation specifications/ drawings		
	6	Recommended list of OB spares, testing tuning spares, tools and test jigs.		
	7	Installation checks, HAT/SAT schedules		
	8	As made drawings including winding & performance data		
	9	Test certificates		
	10	Software version Description giving details for final binary code of all programmable devices in the system including their checksum etc.	Ship staff only shall get the hard copies.	
	11	Final version of the binary (fused) code of all the software in the system on suitable media like CD.		Only Soft copy of manuals shall be distributed to Naval authorities as per the approved INBR Distribution list
	12	Firmware Support Manual. This shall contain the procedure to reprogram all the programmable devices.		
	13	As fitted drawings velographs.		
	Chapter 9 - Quality Assurance			

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9.1	The detail design, material and workmanship shall be in accordance with the best worldwide recognized marine practices, to ensure reliability, durability and ease of maintenance which comply with the ship's requirements. The design shall be such that weight and size are reduced to the minimum practicable, ensuring no compromise in reliability or significant design criteria. A Quality Assurance Program shall be specified by the firm in his offer.
9.2	Quality Assurance Program:
9.2.1	Details of test equipment, test methods, preliminary qualification tests, etc shall be indicated. All tests shall be carried out by firm at his own works, in presence of and inspecting authority, prior to dispatching the equipment to the customer.
9.2.2	Quality assurance plan proposed and any subsequent amendment shall be discussed and agreed upon between IRS & the firm prior to placement of order.
Chapter 10 - Inspection & Testing	
10.1	Inspection: (a) For Design : Shipyard (b) For Installation : Warship Overseeing Team (WOT) of IN at HSL, Vizag (c) For ship trials : IRS (d) For QC : IRS and IHQ MoD(N) (e) For approval of HATs/SATs schedule: IRS
10.1.1	The product offered by the firm shall conform to standard engineering practices. The equipment shall be subject to stage inspection and final test and trials by IRS. Any deviation from the mentioned specifications shall have to be brought to the notice of IHQMOD(N).
10.2	FATs: FAT for the first system shall be carried out at OEM premises by the reps of classification society including QA representative against ATP mutually agreed by IRS and OEM. During FATs firm would demonstrate system shall meet the functional and performance requirements indicated as per approved SOTR. Firm shall submit FAT document to IRS within 03 months after placement of order for approval. The FATs procedure shall comprise of: (a) Functional test (b) Verification of design, especially firmware and software (c) Test equipment used, calibration requirements. (d) Pass / Fail Criteria. (e) Expected duration and time phasing
10.2.1	One copy of FAT report shall be submitted to IHQMOD(N) and shipyard for record/ comments.
10.3	HATs/SATs: HAT/SAT Schedule shall be prepared by firm and forwarded to IRS for Approval within 06 months of Purchase order issued to firm.
10.4	Service of Engineers: The firm shall provide the necessary service of engineer to carry out the following activities: a) Setting to work

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	b) Harbor and Sea Trials
10.4.1	Firm shall depute firm's experienced, qualified and expert engineer/technician to carry out the above activities onboard in-line with ship's construction timeline for the system. Above activities shall be undertaken to ensure satisfactory performance of the system onboard and for proving the system to the inspection agencies as per approved test procedure.
10.4.2	The firm shall indicate total cost (in price bid) for executing all technical activities in the offer. No extra man-days/ payment on this account shall be admissible. Activity-wise cost break up shall also be mentioned in the price bid for reference.
10.5	System Performance responsibility: The firm shall present the system for performance Evaluation Trials (HAT/SAT) to prove the expected performance assured. In case of any irregularities in the operation/ performance of the system or non-conformance to specified parameters observed on integration with ships system, the firm is bound to rectify the defect. The firm shall ensure complete responsibility of satisfactory operation of the system on board.
	Chapter 11 - Other terms and Conditions
11.1	Shipping: All equipment shall be adequately packed and protected with supports to ensure adequate protection during all modes of transportation. Each unit within a package/container shall be clearly marked for identification. The container shall clearly indicate the item description with caution marks, quantity, weight, size, etc. A separate document giving complete details and instructions for storage, preservation, handling and transportation after delivery shall be supplied, The firm shall indicate the delivery schedule, port of embarkation, transport, packing, preservation, insurance, etc.
11.2	Training: The OEM shall conduct an operational and maintenance training course at the shipyard premises to the ship's crew / maintainers. Complete training package for on board exploitation, maintenance and for second level maintenance along with tools and test kit to be offered. The level and number of personnel recommended for training also needs to be indicated and shall be decided during TNC. All training material shall thereafter be handed over to Navy for future training requirements, The training schedule shall be prepared in consultation with WOT/Ship Staff well in advance.
11.3	License: If the equipment is manufactured under license in collaboration in India, the firm is to indicate the following:- (a) Collaboration agreement already signed or not (b) Name and address of the collaborator/principal and details of registration license (c) The extent of indigenous manufacture of components, transfer of technology and collaboration for supply and support of equipment including those supplied directly by the collaborator (d) The yearly production capacity of the Indian licenses

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11.4	Sufficiency Clause: All deliverables supplied with the equipment shall be sufficient for satisfactory operation of the system. If any additional item is thereafter required for satisfactory operation of the system as described in the tender, the same shall be supplied by the firm with no cost implication.
11.5	The cable laying shall be carried out by HSL and the connectorization, STW, HATs and SATs are in Supplier's scope.
11.6	Preservation, Packaging and Shipping: The stores (including OBS and B&D Spares) shall be supplied in long-term preserved condition that is suitable for storage under tropical high humidity conditions for a period of 24 months.
11.7	Distributed UPS complying to EED-50-77 for back up supply shall be included in the technical offer iaw IHQ, MoD(N)/DEE policy letter EE/03/9711/Policy/L-105 dated 24 Jan 17.
11.8	All power cables should comply with EED-50-12 (Rev-2) and all internal cabling should be carried out with LFH cables.

DGM(ED)

M(ED)

CE(ED)

Scope of Supply for MB/SRE System per ship as below:

S.No.	Description of Electrical / Equipment with accessories	Item code	Qty./ Ship
1	IP Based MB/SRE System consisting of		01 Set
1.1	CENTRAL CONTROL UNIT WITH SERVER AND REQUIRED CONTROL CARDS	ICS1001-1881	01 Set
1.2	BOX LOUDSPEAKER 4W(NWT) WITH VOLUME CONTROL	ICS3110	172 Nos
1.3	HORN LOUD SPEAKER (15W) (WT) + VOLUME CONTROL UNIT	ICS3100+ICS3130	110 Nos
1.4	15W (WT) HORN LOUDSPEAKER (BLASTPROOF&DUSTPROOF)	ICS9621+ICS3130	8 Nos
1.5	HORN SPEAKER 50W (WT)	PAN9626	2 Nos
1.6	WALL MOUNTED HANDSET STATION	PAN5051	9 Nos
1.7	HANDSET WITH PLUG	PAN8512	9 Nos
1.8	FLASHER (COTS ITEM)	FLASH-1881	12 Nos
1.9	MULTI WAY JUNCTION BOX	ICS5110	7 Nos
1.10	NETWORK SWITCH 8 PORT WITH ENCLOSURE	SWITCH-1881	4 Nos
1.11	STOWAGE BOX FOR MIC	ICS1510	5 Nos
1.12	UPS	UPS-1881	01 No
1.13	REQUIRED BATTERIES FOR 1 HOUR BACKUP	BATT-1881	01 Set
2	On Board spares 02 years operation shall be as follows:		01 Set
2.1	HANDSET WITH PLUG	PAN8512	01 No.
2.2	BOX LOUDSPEAKER 4W(NWT) WITH VOLUME CONTROL	PAN9625	01 No.
2.3	SPEAKER ELEMENT	SPKR-1881	08 Nos.
2.4	DRIVER UNIT	DRIVER-1881	03 Nos.
3	Drawings & Documentation: Documents shall include User handbook, Technical manual, Servicing and maintenance manual, Installation specs / Drawings, FATs, HATs & SATs.		09 Sets.
3.1	Soft copy manuals in CDs		02 Nos.